

Revised Long-Memory Limit Theorems for the Hurst Function Estimator

Machine-checked against the Lean 4 formalization (repo `hurst_function_estimation`, closure commit `16715db`, acceptance `e76475a`)

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Scope. This note states the *revised* form of the long-memory branch of Theorem 3.3(iii) and Theorem 3.4 of the source manuscript (`19-A0S1825.pdf`), as it is now fully formalized in the accompanying Lean 4 project. Every condition and every clause of Theorem 2 below is machine-checked; §4 maps each element to its Lean theorem, and §5 lists, without softening, every deviation from the manuscript’s original assumptions and from the intermediate repair plan (`direct_proofs` files 22/24).

1 Setup

Let $f \in \Sigma(p, M)$ be a Hölder-regular Hurst profile on $(0, 1)$ in the sense of the source paper, let $t \in (0, 1)$ be a fixed interior point, and write

$$h = f(t), \quad \psi = 2 - 2h \in (0, \tfrac{1}{2}), \quad c = h(2h - 1) > 0.$$

The data are the centered Gaussian grid features of the harmonizable-type model with profile f ; the estimator is the local-polynomial log-square statistic with weights $w_{n,i}$ of order r (degree $r - 1$ design, $\sum_i w_{n,i} = 1$), bandwidth $\delta_n = n^{-\gamma}$, active scale $S_n = n\delta_n$, and the clipped calibration

$$\hat{G}_n = \sum_i w_{n,i} \log(Y_{n,i}^2), \quad \hat{H}_n = T_n \left(\frac{c_\sigma - \hat{G}_n}{2 \log n} \right),$$

where $Y_{n,i}$ are the standardized first-stride increment features (unit known scale $\sigma = 1$) and T_n clips its argument to $[0, 1]$; where T_n clips to $[0, 1]$ and $c_\sigma = \mathbb{E} \log Z^2$ with $Z \sim N(0, 1)$ (the pinned center; in the formalization this constant is `gaussianLogSquareMean`).

Let ω_r be the equivalent kernel of the local polynomial fit and let K be the Hilbert–Schmidt operator of the (unweighted, exponent- ψ , coefficient- c) Riesz kernel on L^2 . Define the compact self-adjoint operator

$$B = K^{1/2} M_{\omega_r} K^{1/2},$$

and let $\lambda = (\lambda_j)_{j \geq 0}$ be its eigenvalue sequence retaining every nonzero eigenvalue with multiplicity, ordered with positive and negative parts separately non-increasing and interleaved.

Condition 1 (Revised assumptions).

(R1) $p \geq 1$; $f \in \Sigma(p, M)$ with constant M .

(R2) $\frac{1}{2} < a \leq f(x) \leq b < 1$ for all $x \in (0, 1)$.

(R3) Long memory at the estimation point: $h = f(t) > \frac{3}{4}$.

(R4) Bandwidth compatibility (the b -band): $(1 - h)^2 < 1 - b$; equivalently, at $\gamma = h$, $(1 - \gamma)\psi < 2 - 2b$.

(R5) Bandwidth pinned at the feasible point $\gamma = h$, i.e. $\delta_n = n^{-h}$ (so $S_n = n^{1-h}$).

(R6) Fixed local-polynomial order $r \geq 0$.

The band (R4) is nonempty jointly with (R2)–(R3): one needs $h \leq b < 1 - (1 - h)^2$, and $h < 1 - (1 - h)^2$ holds for every $h \in (0, 1)$.

2 Main theorem

Theorem 2 (Revised Theorem 3.3(iii) + 3.4, long-memory branch). *Under (R1)–(R6), with Q the centered second Gaussian chaos*

$$Q = \sum_{j \geq 0} \lambda_j (Z_j^2 - 1), \quad (Z_j)_{j \geq 0} \stackrel{iid}{\sim} N(0, 1),$$

(the series converging in L^2 ; $\text{Var } Q = 2 \sum_j \lambda_j^2$), the truth-centered estimator satisfies

$$2 S_n^\psi \log n (\hat{H}_n - h) \implies -Q.$$

The sign is $-Q$: the reflection relative to the naive inversion is a substantive part of the repair (it changes the law whenever Q is asymmetric), and it is retained here exactly as in the corrected mathematics (*direct_proofs/14*).

Corollary 3 (No tuning constants). *Under (R1)–(R4), (R6) alone — with the near/far cut radius and every scalar rate parameter instantiated internally at the midpoint of the admissible window — the conclusion of Theorem 2 holds. The center constant c_σ and the projection band $d = (1 - h)/2$ are likewise internal.*

3 The honest rates behind the theorem

Proposition 4 (Drift decomposition; revised E5). *Under (R1)–(R6) the deterministic drift splits as*

$$2 S_n^\psi \log n |\mathbb{E} \hat{H}_n - h| \leq \underbrace{C_1 \frac{\log n}{n^{h-2(1-h)^2}}}_{\text{local-smoothing term}} + \underbrace{C_2 n^{-(2-2b)-2(1-h)^2}}_{\text{increment-mismatch term}} + \underbrace{C_3 \frac{1}{n^{2(1-h)^2} \log n}}_{\text{clipping term}},$$

where the first exponent is positive since $h > 3/4 > 2(1 - h)^2$, the second is positive exactly under the b -band (R4), and the third vanishes unconditionally. Moreover the increment-mismatch rate $n^{-(2-2b)}$ is attained exactly: no per-row expansion order reaches the rate n^{-h} , and consequently the manuscript's original bias envelope $|\mathbb{E} \hat{H}_n - h| \leq C n^{-h}$ fails for every non-constant profile with $f'(t) \neq 0$ (the window refutations are machine-checked; the lower-bound statement is derived by hand and registered).

Remark 5. Theorem 2 is therefore proved *through* the honest rates: the truth-centered limit needs only the drift to vanish, not an n^{-h} -rate bias envelope, and the drift condition reduces exactly to the b -band (R4).

4 Formal provenance

Every element of Theorem 2 is machine-checked; the chain, from bottom to top:

Element	Lean theorem	File
Internal Hilbert basis, closed spectrum	<code>exists_weightedRieszSpectrum_equiv_closed</code>	<code>WeightedRieszSpectrumClosed</code>
Cycle-integral identification ($k \geq 2$)	<code>diagSum_Bop_eq_weightedCycle</code>	<code>CompressionGeneralK</code>
Signed sorted spectrum keeps moments	<code>signedSortedSpectrum_hasSum_pow</code>	<code>GeneralSignedPowerMatching</code>
Row nondegeneracy (all n ; spectral)	<code>normalizedVaryingIncrement_ne_zero</code>	<code>FeatureRowNondegenerate</code>
Normalized energy $S^{2\psi-2} \sum \rho^2 = O(1)$	<code>hurstHolder_q1_actual_normalized_energy_*</code>	<code>NormalizedActualEnergy</code>
Fourth-order remainder $\rightarrow 0$	<code>hurstHolder_q1_actual_weighted_fourth_tendsto_zero</code>	<code>NormalizedActualRemainder</code>
Quadratic endpoint (constructed law)	<code>actualQ1LongStatistic_tendsto_secondChaos_generalSigned</code>	<code>ActualQ1SignedClosed</code>
Log layer (double-sum remainder)	<code>actualQ1_logStatistic_tendsto_secondChaos_doubleSum</code>	<code>P5LogHLayers</code>
Seam (Layer 1 $\rightarrow$ Layer 2)	<code>logStatistic_knownScaleH_seam_of_logLimit</code>	<code>P5SeamClosed</code>
Center band discharged ($d = (1 - h)/2$)	<code>centerBand_hband_discharged</code>	<code>CenterBandDischarge</code>
Scalar rates ($R = \lfloor n^\beta \rfloor$)	<code>feasibleR_*</code>	<code>FeasibleRates</code>
Second-order increment rate $n^{-(2-2b)}$	<code>varyingIncrement_norm_sq_secondOrder</code>	<code>IncrementLogSecondOrder</code>
Tail-based clipping transfer	<code>truncationCorrection_le</code>	<code>TruncationTransfer</code>
Unit variance rate $S^{-2\psi}$	<code>hurstHolder_q1_unitVariance_eventually_bounded</code>	<code>UnitVarianceRate</code>
Honest-rate lemma	<code>honestRate_drift_tendsto_zero</code>	<code>HonestRateDrift</code>
Main theorem	<code>actualQ1_knownScaleH_fullChain_honestRate</code>	<code>HonestRateClosure</code>
Refuted historical premises (audit)	<code>session3_fullChain_all_rows_impossible, session4_unnormalized_hE2_impossible, bias_window_*_infeasible</code>	<code>verification/</code>

Machine status. Lean 4.31.0 / mathlib v4.31.0; aggregate build 9223 jobs, exit 0; axiom audit of 2603 theorems, all within `{propext, Classical.choice, Quot.sound}`; zero sorry; automated re-verification via `verification/final_acceptance.py` (seven checks, all passed).

5 Deviations from the manuscript and the repair plan

1. **Sign.** The limit is $-Q$, not Q (substantive; matches the corrected mathematics of file 14, confirmed in Lean).
2. **Bandwidth system.** The manuscript’s original bandwidth/all-rows combination is unsatisfiable; both refutations are machine-checked. The revised window is (R4)–(R5), exactly the file-22 grid condition $(1 - \gamma)\psi < 2 - 2b$ instantiated at $\gamma = h$.
3. **Lower bound on a .** (R2) requires $a > \frac{1}{2}$, absent from the manuscript; it is needed by the second-order mixed-power machinery. Mild strengthening, satisfiable.

4. **γ pinned.** The honest-rate closure is assembled at the single feasible point $\gamma = h$; the manuscript’s window allows other γ (the general- γ chain exists in the repo with a legacy bias premise). Coverage regions overlap but neither contains the other.
5. **Scale.** Known scale pinned at $\sigma = 1$ ($c_\sigma = \mathbb{E} \log Z^2$); the σ -rescaled known-scale variant and the unknown-scale extension (file 24, E6–E10) are not formalized.
6. **Honest bias rates.** The drift uses the attained rate $n^{-(2-2b)}$ (Proposition 4); the plan’s presumed faster decay ($n^{-(2-b)}$, then $n^{-\gamma}$ envelopes) was refuted during formalization and is registered, not papered over.
7. **Short-memory and critical branches** of Theorem 3.3, the minimax program, and all two-dimensional extensions are outside this closure and remain as separately registered work.

6 Validation status of the remaining results

Theorem 2 closes the long-memory branch. The repository audit (`results/catalog.json`; 27 numbered results; 2,644 machine-checked mathematical declarations) grades every numbered result of the manuscript; the full per-item detail with Lean theorem names is maintained in `paper_revision/theorem_inventory.md`. We summarize.

Status tags. **[Full]**: original result completely proved in Lean (repaired form). **[Revised-Full]**: revised statement completely proved in Lean. **[Cond.]**: conditional Lean theorem under named premises (INT-CLT, INT-VAR, INT-LONG, INT-SCALE, INT-MOM-RISK). **[Written]**: ordinary mathematical proof only (files 1–24). **[Open]**: no conclusion yet. “[Full pieces]” marks substantial unconditional Lean fragments inside a partially open item.

Result	Claim (one line)	Status
3.1	minimax L^s probability-risk lower bound, $d = 1$	[Full], incl. $p = 1$ extension
3.2	local-poly bias $O(T_1)$ /variance $O(T_2)$ on Ω_δ	short-memory global bounds [Full pieces]; uniformity [Written]
3.3	short/critical/long-memory limits of $\widehat{G}_n$	short, critical [Cond.]; long-memory [Revised-Full] (Thm. 2)
3.4	inverted $\widehat{H}_n$: bias, MSE, limit law	long-memory inversion [Revised-Full]; rest [Cond.]/[Written]
3.5	optimal-bandwidth biased normal limits	means corrected to $-2R/2R$; [Cond.]
4.1	two-scale pilot $\widehat{H}_1$	[Cond.]
4.2	$\log \sigma^2$ estimator MSE	[Written]; $q = 1$ scale input discharged in Lean
4.3	plugged-in $\widehat{H}_2$ MSE	fixed-class minimax match [Full pieces, $1 \leq s \leq 2$]; rest [Written]
8.1	KL bound for $H = \frac{1}{2} + \varphi$	[Full]
8.2	four difference-covariance regimes	growing-lag remainder corrected; interior [Written + Full pieces]
8.3	log-mean chain $E[2 \log W_n]$	actual full-grid means [Full pieces]
8.4	local-poly bias $O(b^p)$	design-matrix repair; designs/weights/equiv. kernel [Full pieces]
8.5	vector Z limits and bounds	centering fixed; expectation-bias leading terms [Full pieces]

Result	Claim (one line)	Status
S.1.1	Fourier adjacent-difference bounds	false as complex integral; repaired version [Open]
S.1.2	filtered Fourier integral bound	route only; [Open]
S.2.1	$g(H, 0, h)$ bounded away from 0	hypotheses strengthened ($h \neq 0$, $H \in [\gamma, 1 - \gamma]$); 1-d [Full pieces]
S.2.2	leading singular coefficient	missing $\frac{1}{2}$ in c_h ; general orders [Open]
S.2.3	mixed-derivative bounds	linear- H counterexample; repaired with r^ψ background [Written + Full pieces]
S.3.1	far-field asymptotics of g	route only; [Open]
S.3.2	weighted power double sums, cycle products	critical center value fixed ($\int f^2$); long-memory cycle integrals [Revised-Full]
S.3.3	three-regime unified bounds	actual row sums/variance [Full pieces]; $s > 2$ [Cond.]
S.3.4	Hermite orthogonality identity	correct as stated; general (k, l) [Open]
S.5.1	spatial-average variance of I'_1	missing m factor ($H = 7/8$ counterexample); repaired [Written]
S.5.2	truncated component I'_2 MSE	[Written + Full pieces]
S.6.1	irregular-grid covariance	false unnormalized; Brownian counterexample Lean-verified; full instantiation [Open]
S.7.1	known $\sigma(t)$ covariances	[Open]
S.7.2	$d \leq 3$ bias/variance, optimality	[Open]; the $d = 2, 3$ optimality claim lacks its own lower bound

6.1 Confirmed errors in the original proofs

(1) Critical variance constant: $(9/16) \int \omega^2$, not the kernel center value (S.3.2). (2) Long-memory sign: the inverted limit is $-Q$; a legal nonnegative equivalent-kernel instance has third moments of opposite sign (§1–2 here; machine-checked). (3) Theorem 3.5 means are $-2R(\hat{G})$ and $2R(\hat{H})$, not R . (4) The growing-lag remainder of 8.2(iv) is too strong; $O(k^{-2} + k \log n/n + (k/n)^\psi)$ must be retained. (5) The quadrature argument of 8.4 is invalid; replaced by direct design-matrix reproduction (Lean-complete). (6) 8.5 mixes Z and $Z - EZ$ centerings; the dimension is S , not d . (7) S.1.1 fails as a complex Lebesgue integral near 0. (8) S.2.1 lacks hypotheses ($h = 0$; $q = 2$, $H \rightarrow 1$). (9) S.2.3's $q = 1$ near-diagonal remainder has a linear- H counterexample. (10) The c_h formulas of S.2.2/S.3.1 miss a factor $\frac{1}{2}$. (11) S.3.2's critical center-value derivation and the $2!$ Hermite factor. (12) S.5.1's spatial-average bound misses an m factor (power-order counterexample at $H = 7/8$; the repaired bound preserves the main rates). (13) S.6.1's unnormalized covariance claim fails already for Brownian motion (counterexample machine-checked at the scalar level). (14) S.7.2's $d = 2, 3$ optimality assertion has no corresponding lower bound in the manuscript itself.

6.2 Refutations formalized during verification

The verification chain itself formalized six impossibility results about the proof infrastructure: the all-rows-nonempty premise (**hm**), the spectral constant (**hconst**) and the unnormalized correlation-energy premise (**hE2**) are each refuted by Lean contradiction theorems; the first- and second-order bias windows are empty at $\gamma = f(t)$ on the long-memory band; the second-order increment rate $n^{-(2-b)}$ is unattainable (honest rate $n^{-(2-2b)}$, Proposition 4); and feeding the crude fluctuation

bound into the drift yields a non-vanishing $O(1)$ term (the tail-based transfer is the only correct route). Four lemmas that looked problematic were positively validated, most notably the spectral row-nondegeneracy (`normalizedVaryingIncrement_ne_zero`), which is stronger than the plan anticipated.

6.3 Open scopes

Dimension $d \geq 2$ throughout; boundary uniformity (Ω_δ); $s = \infty$ and the $s > 2$ moment bridge (INT-MOM-RISK); the unknown-scale fine limits (file 21, written only); the INT-CLT/INT-VAR inputs of the short/critical branches; and the eight supplement items marked [Open] above.